

Spike: 7**Title:** Agent Marksmanship**Author:** Edward Herrod, 104541862**Goals / deliverables:**

The goal of this spike is to create a working simulation of an agent which accounts for a target's movements when aiming a weapon. Weapons themselves are variable and can be changed with a key bind (Q/E). Where the agent is aiming is represented visually.

Technologies, Tools, and Resources used:

List of information needed by someone trying to reproduce this work:

- Visual Studio Code
- Python 3.x
- Pyglet 2.1.x
- Git
- Github (for installation of code)

Tasks undertaken:

- Download and install Visual Studio Code
- Download and install the latest version of Python
- Install the latest version of Pyglet (pip install --upgrade --user pyglet)
- Import the lab containing the base code for this spike (Task 12)
- Create a TargetAgent class which serves as the moving target
- Create aiming logic within the Agent class and instantiate an agent into the game world
- Create a Weapon class with a WEAPON_STATS dictionary for all weapon parameters for each type (color, length, width, projectile_speed etc)
- Ensure the weapon points where the agent is aiming
- Create a Projectile class which can be explosive or not and ensure that each one is instantiated from the muzzle of the weapon
- Add firing logic – for this spike, the weapon fires on a left mouse click
- Have the target visually register hits by giving it collision detection with projectiles

What we found out:

Figure 1: The agent aims at a moving target – the weapon points to where the target is going, not where it is, dependent on how fast the bullets travel.

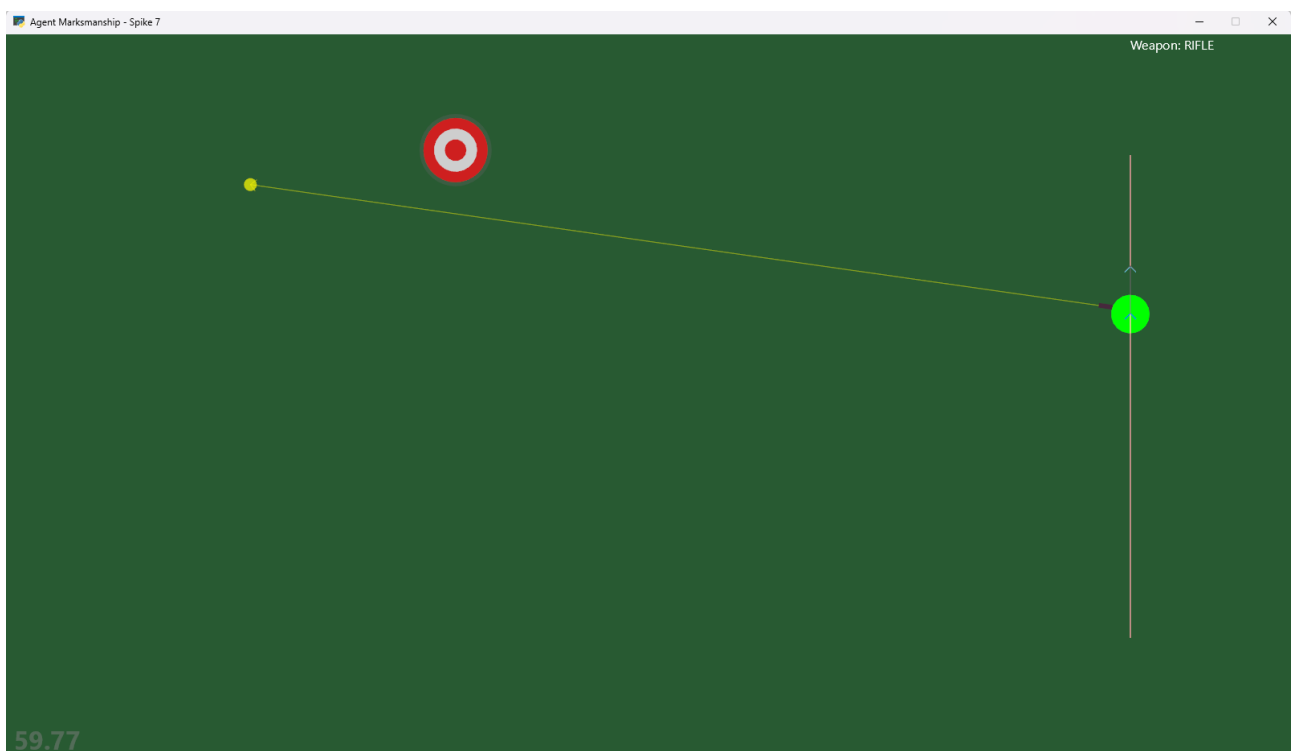


Figure 2: Visual demonstration of where the agent is aiming. The target's collision circle is highlighted in grey and is slightly larger than the target itself. The agent patrols the line in pink and the target moves around randomly.

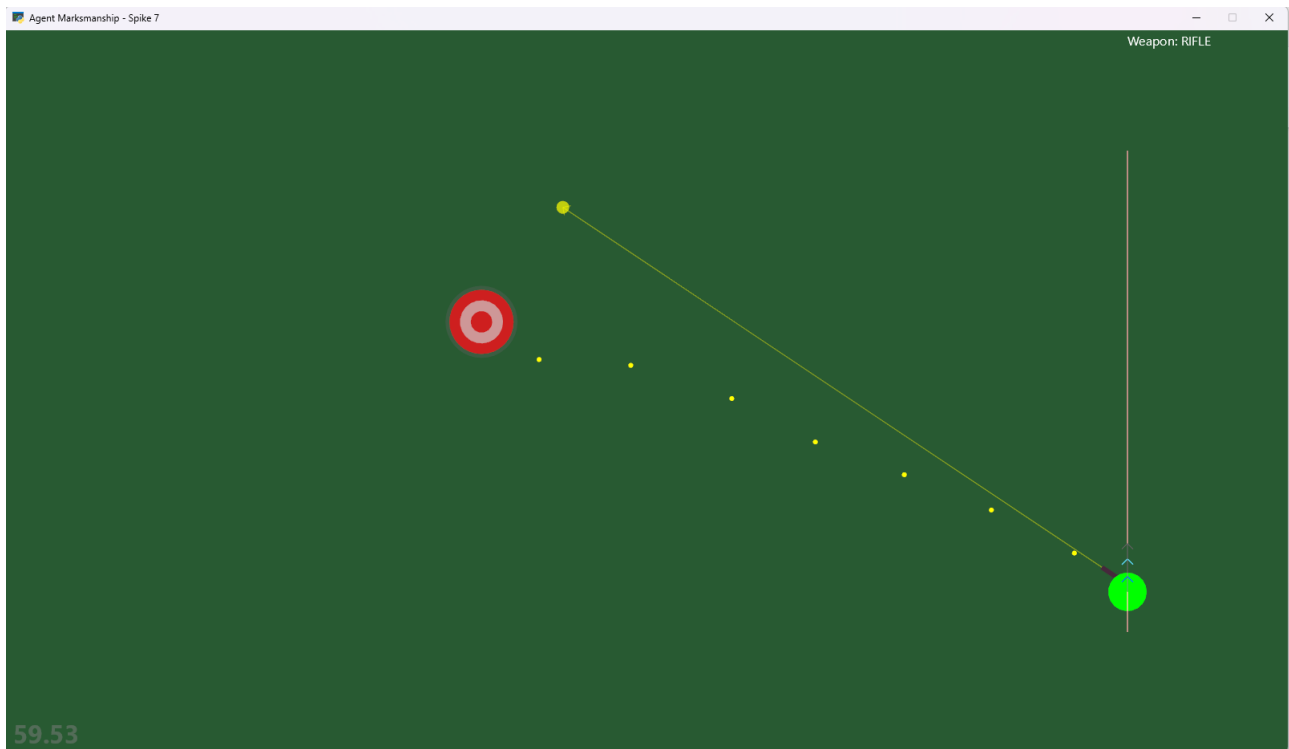


Figure 3: The target's white ring flashes pink when hit to visually indicate bullet collisions.



Figure 4: When the agent is holding the shotgun, it switches from patrolling the pink line to chasing the target, maintaining separation when possible. The shotgun shoots multiple projectiles at a short range.

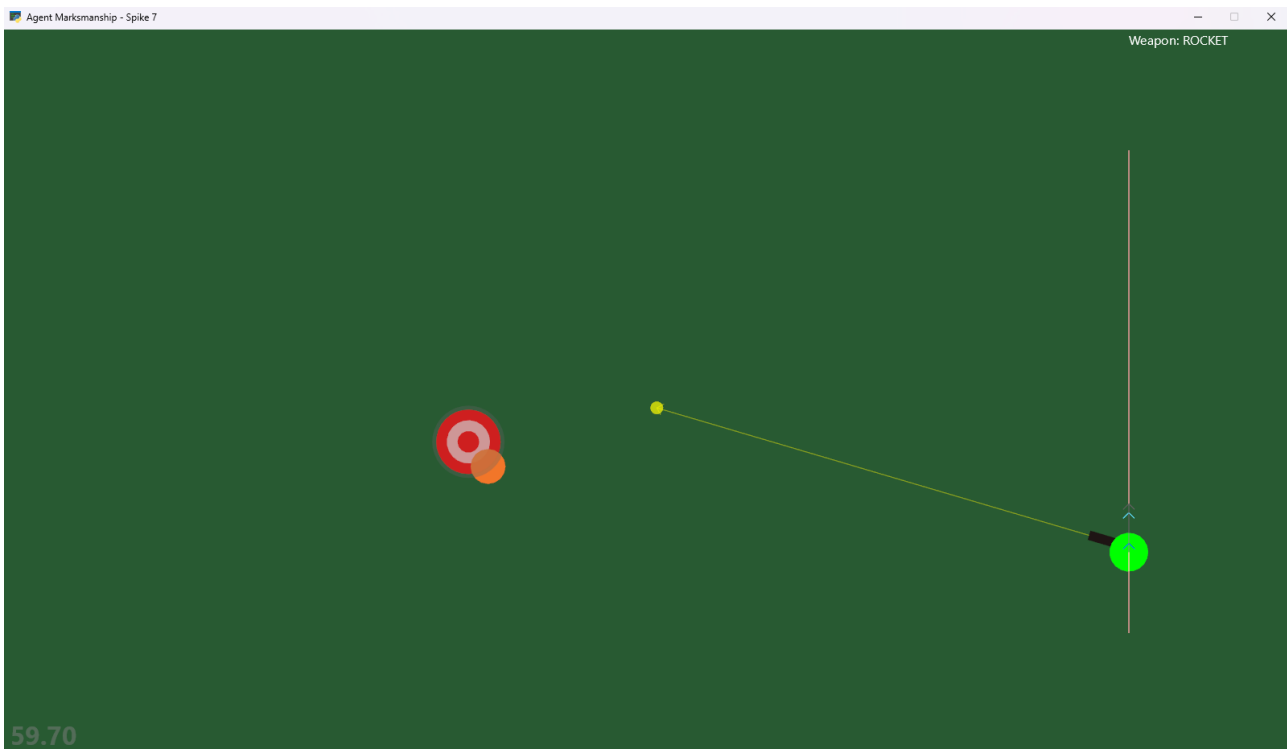


Figure 5: The resultant explosion from a rocket projectile hitting the target. Note how distant the predicted aiming position is from the target due to the rocket's slower movement.